

Multi-Agent Financial Research System

UI/UX Design Specification — Volume 1 of 4

Foundation: Product Vision · UX Strategy · Information Architecture · Design System · Navigation · Component Library

Prepared for: Production build via Stitch (Google AI UI Designer) **Prepared by:** Senior Software Development & UI/UX Design Function **Version:** 1.0 · Status: Production-Ready Specification

How to Use These Volumes in Stitch

Feed each section as a design brief in order. Volume 1 establishes the visual language and component vocabulary that every screen in Volumes 2–4 references by name (e.g. "Primary Button," "Citation Pill," "Agent Status Chip"). Keep Volume 1 open as your style-guide truth source while generating individual screens so Stitch stays consistent across all ~40 screens.

1. Product Vision

1.1 Vision Statement

A financial research workspace that feels like sitting next to a team of six expert analysts who never sleep, never guess, and always show their work. Every number on screen is clickable back to its source page. The product should feel **precise, calm, and intelligent** — never flashy, never chatty, never uncertain.

1.2 Design North Star

"An institutional-grade Bloomberg Terminal crossed with the clarity of Linear and the warmth of Notion — made legible for a finance student and trustworthy for a CFA."

1.3 UX Goals (ranked)

1. **Trust through transparency** — every AI claim visibly traces to a page citation; nothing "feels" like a black box.
2. **Speed to insight** — from upload to first answered question in under 3 minutes, with zero dead-ends.
3. **Cognitive clarity under density** — financial data is dense; layout must reduce, not add, cognitive load (generous whitespace, strict hierarchy, one primary action per screen).

4. **Confidence in AI activity** — users should always know which of the 6 agents is working, why, and what it produced (never a silent spinner).
5. **Delight without distraction** — 3D and motion reinforce meaning (e.g., a document "becoming" structured data) rather than decorate for its own sake.

1.4 Design Principles

Principle	What it means in this product
Show, don't assert	Every metric has a "source" affordance (page badge) instead of a disclaimer footnote.
One agent, one voice	Each of the 6 agents has a distinct icon/color accent so users build a mental model of "who" produced what.
Progressive disclosure	Dashboards show summaries first; drill-down reveals raw citations, chunk text, and confidence.
Never a naked wait	Every async action (upload, extraction, report generation) has a purpose-built loading state that narrates progress, not a generic spinner.
Reversible, forgiving	Destructive actions (delete workspace/document) always confirm; nothing auto-deletes silently.

1.5 Primary Personas (from PRD) — UX Implications

Persona	UX Implication
Priya , Finance Student	Needs guided empty states, tooltips on financial terms, a fast "ask a question" path front and center.
Rohan , MBA Candidate	Needs Comparison mode to be a first-class nav item, not buried; side-by-side visual emphasis.
Ananya , Early-Career Analyst	Needs Red Flags surfaced proactively on the dashboard, and a one-click path to a shareable report.
Admin/Developer (internal)	Needs an Agent Activity Monitor and logs view, kept out of the primary end-user nav (separate developer area).

2. Information Architecture

2.1 Top-Level Site Map



- | | | — Preferences
- | | | — Billing (future-scope placeholder)
- | | — Help Center (in-app)
- |
- | — System States (cross-cutting, overlay on any route)
- | | — Error Pages (404 / 500 / 403 / offline)
- | | — Loading Screens (route-level + component-level)
- | | — Empty States (per-module)
- |
- | — Admin / Developer (internal, gated route)
- | | — Agent Logs Explorer
- | | — Seed Document Manager
- | | — System Health

2.2 Navigation Depth Rule

Maximum 3 clicks from Dashboard to any financial insight: **Dashboard → Workspace → Tab (Chat/Metrics/Comparison/Red Flags)**. Reports and Settings are exceptions at depth 2 via global nav.

2.3 Screen Inventory (Master List — 42 Screens)

This is the authoritative list; every screen below is specified fully across Volumes 2-4.

#	Screen	Volume
1	Landing Page	2
2	Login	2
3	Register	2
4	OAuth (Google/GitHub) Bridge Screen	2
5	Forgot / Reset Password	2
6	Onboarding — Welcome Carousel	2
7	Onboarding — First Workspace Setup	2
8	Dashboard (Home)	2
9	Workspace List	2
10	Create Workspace (Stepper Modal)	2
11	Workspace Overview (shell + tabs)	2
12	Upload Documents	2
13	Document Processing (live status)	2
14	Document Library	2
15	AI Chat	3
16	Chat History Drawer	3
17	Citations Viewer (panel + modal variant)	3
18	Financial Metrics Dashboard	3
19	Company Comparison	3
20	Red Flag Analysis	3
21	Agent Activity Monitor	3
22	Report Generation Flow (3 steps)	4
23	PDF Preview	4

#	Screen	Volume
24	Reports Library	4
25	Search (global overlay)	4
26	Notifications Drawer	4
27	Profile	4
28	Settings — Account	4
29	Settings — Security	4
30	Settings — Preferences	4
31	Help Center	4
32	Error — 404 Not Found	4
33	Error — 500 Server Error	4
34	Error — 403 Forbidden	4
35	Error — Offline	4
36	Loading — Route Transition	4
37	Empty States (composite spec, 6 variants)	4
38	Mobile Variants (composite spec, key screens)	4
39	Admin — Agent Logs Explorer	4
40	Admin — Seed Document Manager	4
41	Admin — System Health	4
42	Toast / Inline Notification System (composite spec)	4

3. Navigation System

3.1 Desktop Navigation Structure

- **Left Sidebar (persistent, collapsible to icon-rail):**
 - Logo / workspace switcher (top)

- Dashboard
- Workspaces
- Reports
- Divider
- Search (opens overlay, also )
- Notifications (badge count)
- Divider
- Help Center
- Settings
- Profile avatar (bottom, opens menu: Profile / Settings / Log out)
- **Top Bar (contextual, within a Workspace):**
 - Workspace name + switcher chevron
 - Tab strip: Chat · Documents · Metrics · Comparison · Red Flags · Agent Activity · Reports
 - Right-aligned: "Generate Report" primary button + Upload icon button
- **Breadcrumb Rule:** Shown only two levels deep — Workspaces / [Workspace Name] — never deeper, since tabs replace deep breadcrumbs.

3.2 Mobile Navigation Structure

- **Bottom Tab Bar (5 slots max):** Dashboard · Workspaces · Chat (workspace-context aware) · Reports · Menu (overflow: Search, Notifications, Settings, Profile, Help).
- Workspace tabs (Chat/Metrics/Comparison/Red Flags/Agent Activity) become a horizontal swipeable segmented control directly under the workspace header, not the bottom bar.

3.3 Navigation States

State	Treatment
Active nav item	Filled icon + accent-colored left border (4px) + label weight 600
Hover (desktop)	Background tint Neutral-100 , 120ms ease
Disabled (e.g., Reports before any doc processed)	40% opacity, tooltip "Upload a document to unlock reports"
Notification badge	Small filled circle, Accent-Red, count capped at "9+"

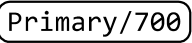
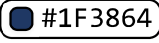
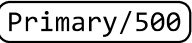
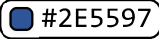
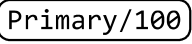
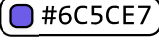
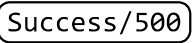
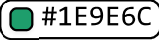
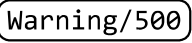
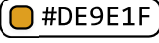
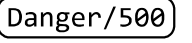
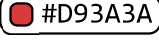
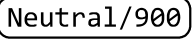
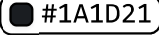
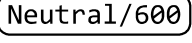
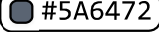
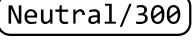
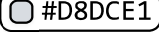
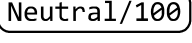
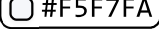
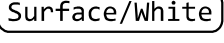
3.4 Keyboard & Power-User Navigation

-   — global search overlay
-  then  — go to Dashboard;  then  — go to Workspaces (Linear-style go-to shortcuts)
-   — send chat message
-  — close any modal/drawer/panel

4. Design System

4.1 Color System

Brand & Semantic Palette

Token	Hex	Usage
 Primary/700	 #1F3864	Headers, primary nav active state, key headings
 Primary/500	 #2E5597	Primary buttons, links, active tab underline
 Primary/100	 #DCE6F1	Selected row/card backgrounds, subtle highlights
 Accent/Violet	 #6C5CE7	AI/agent-related accents (chat bubbles, agent avatars) — reserved exclusively for "AI is speaking/acting" moments
 Success/500	 #1E9E6C	Positive metrics, "no red flags," upload success
 Warning/500	 #DE9E1F	Medium-severity red flags, caution banners
 Danger/500	 #D93A3A	High-severity red flags, destructive actions, errors
 Neutral/900	 #1A1D21	Primary text
 Neutral/600	 #5A6472	Secondary text, captions
 Neutral/300	 #D8DCE1	Borders, dividers
 Neutral/100	 #F5F7FA	App background
 Surface/White	 #FFFFFF	Cards, panels, modals

Data Visualization Palette (Comparison charts, metrics) #2E5597 (Company A) · #DE9E1F (Company B) · #1E9E6C (Company C) · #6C5CE7 (Company D) — colorblind-safe, tested against Deuteranopia/Protanopia simulation.

Agent Identity Colors (used consistently across avatars, status chips, log entries)

Agent	Color
Document Agent	#4C6EF5 (blue)
Extraction Agent	#12B886 (teal-green)
Red Flag Agent	#E8590C (orange-red)
Comparison Agent	#7048E8 (violet)
Research Agent	#1971C2 (deep blue)
Report Agent	#0CA678 (emerald)

4.2 Typography

Typeface: Inter (UI text) + Lora (optional, for report/PDF document contexts only, to feel "document-like").

Style	Size / Line-height	Weight	Usage
Display	36px / 44px	700	Landing page hero only
H1	28px / 36px	700	Screen titles
H2	22px / 30px	600	Section headers
H3	18px / 26px	600	Card/subsection headers
Body Large	16px / 24px	400	Chat answers, primary reading content
Body	14px / 20px	400	Default UI text
Caption	12px / 16px	400	Metadata, timestamps, citations
Numeric/Mono	14–20px	600	Financial figures (tabular-nums enabled) — uses IBM Plex Mono for perfect digit alignment in tables

4.3 Spacing & Layout Grid

- **Base unit:** 8px. All padding/margin in multiples of 8 (4px allowed for micro-adjustments only).
- **Desktop grid:** 12-column, 24px gutter, max content width 1440px, side margins fluid ≥ 32px.
- **Breakpoints:** sm 360-767px · md 768-1023px · lg 1024-1439px · xl 1440px+.
- **Card padding:** 24px standard, 16px in dense tables/comparison grids.

4.4 Elevation & Depth

Level	Shadow	Usage
E0	none	Flat page background
E1	0 1px 2px rgba(16,24,32,0.06)	Cards at rest
E2	0 4px 12px rgba(16,24,32,0.10)	Hover cards, dropdowns
E3	0 12px 32px rgba(16,24,32,0.16)	Modals, drawers
E4	0 24px 64px rgba(16,24,32,0.24)	3D hero elements, onboarding illustrations

4.5 Corner Radius

4px inputs/chips · 8px buttons/cards · 16px modals · 999px pills/avatars/badges.

4.6 Iconography

- **Icon set:** Phosphor Icons (duotone variant for empty states/onboarding, regular weight for UI).
- **Sizes:** 16px (inline text), 20px (buttons/nav), 24px (headers), 48px+ (empty state illustrations).
- Agent icons are custom-drawn (see 3D Visual Language) rather than generic Phosphor icons, to build agent identity.

4.7 3D Visual Language

3D is used **purposefully**, not decoratively — reserved for four contexts:

1. **Onboarding & Empty States** — a soft, isometric 3D illustration of a document "flowing" into structured blocks (representing parsing → chunking → embedding),

rendered in a low-poly, pastel-toned style consistent with the brand palette. Used on: Welcome Carousel, empty Document Library, empty Chat.

- 2. **Agent Avatars** — each of the six agents gets a small 3D "badge" icon (soft-clay / claymorphism style, subtle bevel and ambient occlusion, NOT photorealistic) in its identity color, used in: Agent Activity Monitor, chat message attribution, processing status.
- 3. **Upload & Processing Visualization** — a 3D animated "document stack → particles → vector nodes" sequence during Document Processing, visually explaining chunking/embedding to non-technical users without technical jargon on screen.
- 4. **Landing Page Hero** — a single, restrained 3D scene: a stylized floating "report" document surrounded by 6 orbiting agent badges, subtly rotating (slow, ambient, non-distracting — see Motion Guidelines, Volume 4).

3D Style Rules:

- Palette restricted to the brand palette + soft neutral gradients — never fully photoreal, never stock-3D-asset look.
- Consistent light source: top-left, soft shadow, no harsh specular highlights.
- All 3D assets exported as Lottie/WebM/optimized GLTF with a static PNG fallback for reduced-motion users and slow connections.
- 3D never blocks or delays functional UI — it is supplementary illustration only, never load-bearing for comprehension.

4.8 Motion Foundations (detailed in Volume 4)

Standard easing: `cubic-bezier(0.4, 0, 0.2, 1)`. Standard durations: micro 120ms, standard 200–240ms, panel/modal 280ms, celebratory/3D 600–900ms.

5. Component Library

Each component below is the canonical spec referenced by name in Volumes 2–4.

5.1 Buttons

Variant	Usage	Spec
Primary	One per screen, main CTA (e.g., "Generate Report")	Filled <code>Primary/500</code> , white text, 8px radius, 40px height, hover darkens 8%
Secondary	Supporting actions	Outline <code>Neutral/300</code> border, <code>Neutral/900</code> text

Variant	Usage	Spec
Ghost	Tertiary/inline actions	No border/fill, Primary/500 text
Destructive	Delete workspace/document	Filled Danger/500
Icon Button	Toolbar actions	32×32px, ghost by default, tooltip required
AI Action Button	"Ask AI," "Regenerate"	Filled Accent/Violet , subtle glow on hover (reserved for AI-triggered actions only)

5.2 Inputs

- Text field, textarea, select, file dropzone, search bar (with ⌕ hint), password field (with visibility toggle + strength meter on Register).
- All inputs: 4px radius, 1px Neutral/300 border, Primary/500 border + subtle focus ring on focus, error state = Danger/500 border + inline caption below.

5.3 Cards

- **Metric Card** — label, large numeric value (mono font), trend arrow, sparkline (optional), source-page badge in top-right corner.
- **Workspace Card** — name, last updated, document count thumbnail stack, red-flag severity dot.
- **Document Card** — filename, page count, processing status chip, thumbnail.
- **Agent Output Card** — agent badge, timestamp, summary text, "View Details" link.

5.4 Chat Components

- **User message bubble** — right-aligned, Primary/100 background.
- **AI answer bubble** — left-aligned, white surface, agent avatar (Research Agent), inline **Citation Pills** appended to sentences.
- **Citation Pill** — small rounded chip, format [DocName · p.42], click opens Citations Viewer panel scrolled to that page.
- **"Thinking" state** — animated 3-dot pulse + subtle agent-badge shimmer (never a generic spinner).
- **No-answer state** — distinct muted-tone bubble: "I couldn't find this in the uploaded documents" with a "Rephrase" suggestion chip.

5.5 Status & Feedback

- **Agent Status Chip** — dot + label: Idle (grey) · Running (agent color, pulsing) ·

Complete (green check) · **Failed** (red, retry icon).

- **Severity Badge** (Red Flags) — **Low** (amber outline) · **Medium** (amber filled) · **High** (red filled, exclamation icon).
- **Toast** — bottom-right, 4 variants (success/info/warning/error), auto-dismiss 5s, manual close always available.
- **Progress Bar** — determinate (upload %, report generation) with label; indeterminate variant only for sub-1s operations.
- **Skeleton Loader** — shimmer placeholders matching the exact shape of the content they precede (never generic grey boxes disconnected from layout).

5.6 Data Display

- **Comparison Table** — sticky first column (metric name), horizontally scrollable company columns, best/worst-performer cell highlighting.
- **Financial Metric Table Row** — label · value (mono) · YoY delta chip · source-page badge.
- **Tabs** — underline style, active = **Primary/500** underline + bold label.
- **Stepper** — used in Create Workspace and Report Generation flows; numbered circles, connecting line, completed = filled check.

5.7 Overlays

- **Modal** — E3 elevation, 16px radius, max-width 560px (small) / 800px (large, e.g. Report Preview).
- **Drawer** — slides from right, used for Chat History, Notifications, Citations Viewer (panel variant).
- **Tooltip** — dark surface, 4px radius, appears on 400ms hover delay, used heavily for financial term definitions.

5.8 Navigation Components

- Sidebar (desktop), Bottom Tab Bar (mobile), Top Tab Strip (workspace context), Breadcrumb (2-level max), Command Palette ( search overlay with fuzzy match across workspaces/documents/reports).

End of Volume 1. Continue to Volume 2: Authentication, Dashboard, Workspace, Document Management, Upload Flow.